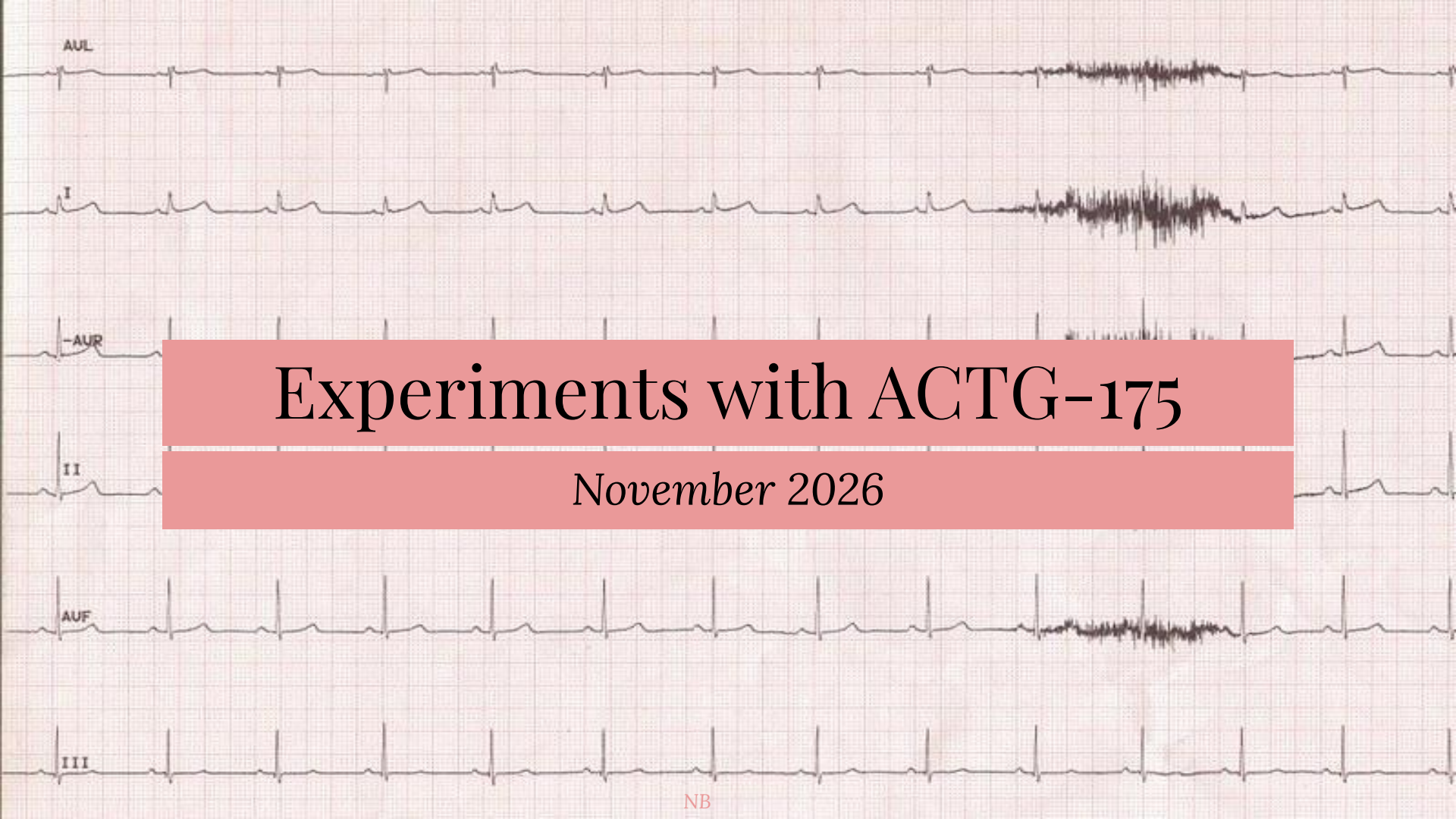


Experiments with ACTG-175

November 2026

Datasets

- The **AIDS Clinical Trial Group 175 (ACTG-175)** dataset is publically available
- The *treat* variable shows whether a patient underwent **zidovudine (ZDV)** monotherapy or not.
  The *arms* variable further breaks it down into ZDV monotherapy and the three increasing doses of **Didanosine (ddl)**

Arm #	0	1	2	3
Arm	**ZDV 600**	**ZDV (600mg) + dd (400mg)**	**ZDV (600mg) + ZAL (2.25mg)**	**dd (400mg)**
Patient Count	532	522	524	561

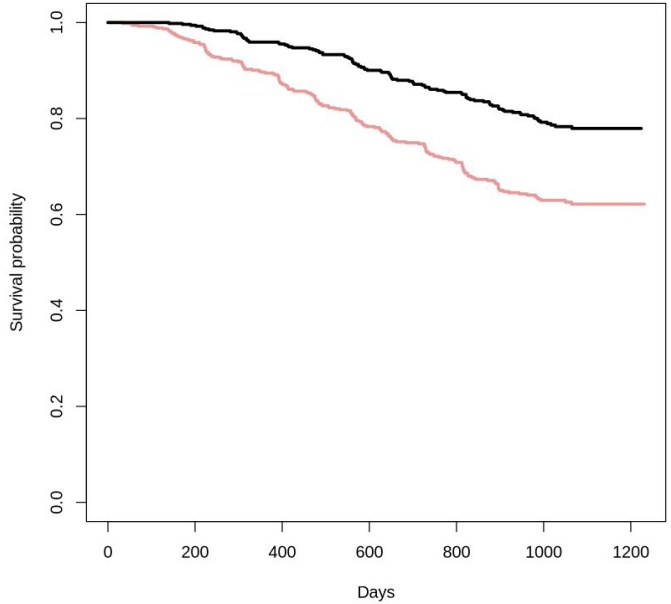
Dataset of interest

NB

Survival analysis

Survival analysis (Treated v. untreated)

- Here, **treated *means* additional dd 200mg on top of ZDV 600mg**
- Dataset of 1054 patients
- Some variables of note:
  - Treated or untreated
  - Weight at baseline
  - Gender
  - Age
  - Karnof scale - ability to tolerate treatments
  - History of IV drug use
- [*Pictured:*] Kaplan Meier curve showing different risks for treated and untreated patients



NB

Survival analysis (Cox)

- Cox PH test : p-value of 0.166 when fit on the above
  i.e. satisfied the Proportional Hazards assumption

- Of the selected variables, treatment had a significant impact, so did response to treatment (karnof as proxy)

- i.e.
  "Both treatment and ability to tolerate harsher treatments reduced risk of death
  With treatment showing a 50% reduced hazard"

Variable	HR (95% CI)	p.value
treat	0.50 (0.39–0.64)	<0.001
wtkg	1.01 (1.00–1.02)	0.1686
gender	1.34 (0.94–1.92)	0.1046
age	1.00 (0.99–1.02)	0.5379
karnof	0.98 (0.96–1.00)	0.0229
drugs	0.68 (0.45–1.04)	0.0728

NB

Survival analysis (Cox) - interaction

- An obvious follow up was to ask if there was an interaction between treatment and ability to tolerate such treatments
- There was no additional impact of being a tolerant treated patient
- So, is there some high yield / high responder group that got treated?
  Causal forests are useful here

Variable	HR (95% CI)	p.value
treat	4.62 (0.11–200.96)	0.4267
karnof	0.99 (0.96–1.01)	0.2508
wtkg	1.01 (1.00–1.02)	0.1725
gender	1.36 (0.95–1.95)	0.0882
age	1.00 (0.99–1.02)	0.5307
drugs	0.68 (0.45–1.04)	0.0761
treat:karnof	0.98 (0.94–1.02)	0.2483

Causal Forests

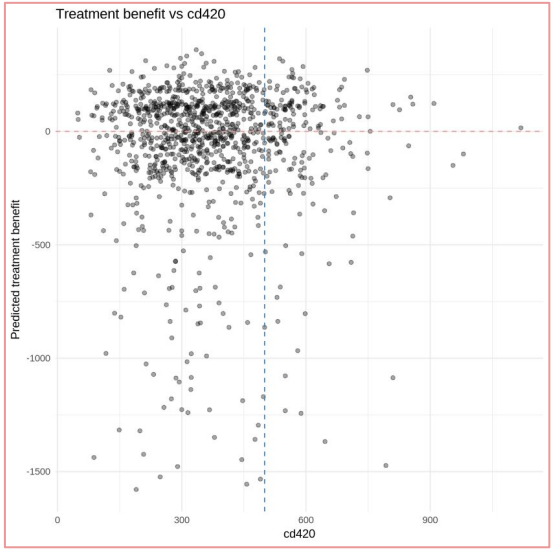
Why a causal forest?

- A causal forest estimates a value called CATE (Conditional Average Treatment Effect)
- Average because it takes patients within a group (say, white 80 year-old men with diabetes, and compares the mean of the treated and untreated members who belong to this group)
- Clinical trials are especially well suited to this, since the patients are randomized for much this reason.
- Since we're explicitly working with survival data, we'll be using the causal *survival* forest specifically.

What does the high-benefit group look like?

When the model was fit on just **'IV drug use history', 'karnof score', and 'weight in kg'**

	Treated	**Untreated**
Zero/ Neg. Benefit	267	255
Positive Benefit	265	267

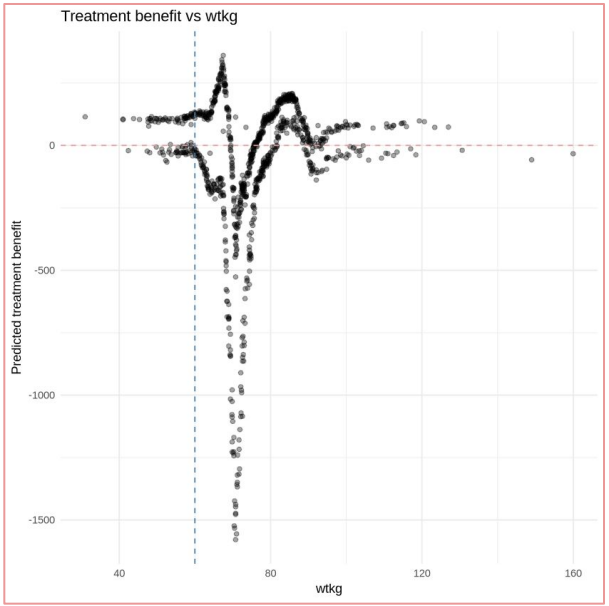


NB

Which variables mattered most in that fit?

When the model was fit on just **'IV drug use history', 'karnof score', and 'weight in kg'**

Variable	**Importance**
Wt. kg	0.734
Karnof	0.264
Drug use	0.001



NB

Now, with more variables:

'cd40', 'symptom', 'karnof', 'age', 'race', 'preanti', 'offtrt', 'wtkg'

- **CD40** gives us their : CD4 T-cell count within 40 days | Not quite baseline, but an early param.
- **Symptom** shows whether they're symptomatic or not
- **Age, Race, Weight** as demographic/descriptive
- **Pre-anti -** how many days prior were they on anti-retroviral therapy
- **Offtrt -** Did they eventually stop treatment

Result:
CD40, Pre-anti and age were most influential in the model

variable	importance
cd40	0.32885677
preanti	0.21950476
age	0.20471487
wtkg	0.16370533
karnof	0.03090401
offtrt	0.02678873
symptom	0.01290423
race	0.01262130

NB

Now, with more variables:

Fitting on Cox fails the Cox PH if cd40 is included

term	HR (95% CI)	p.value
cd40	1.00 (1.00–1.00)	<0.001
symptom	1.37 (1.04–1.80)	0.0258
karnof	0.99 (0.97–1.01)	0.2284
age	1.00 (0.99–1.01)	0.9567
race	0.83 (0.62–1.10)	0.1943
preanti	1.00 (1.00–1.00)	<0.001
offtrt	1.78 (1.40–2.28)	<0.001
wtkg	1.01 (1.00–1.02)	0.0150

NB

Now, with more variables:

Passing the Cox PH assumption:

*Fitting on Cox fails the Cox PH if offtrt is included

term	HR (95% CI)	p.value
cd40	1.00 (1.00–1.00)	<0.001
symptom	1.39 (1.06–1.83)	0.0190
karnof	0.98 (0.96–1.00)	0.0768
age	1.00 (0.98–1.01)	0.6947
race	0.81 (0.61–1.08)	0.1561
preanti	1.00 (1.00–1.00)	<0.001

NB

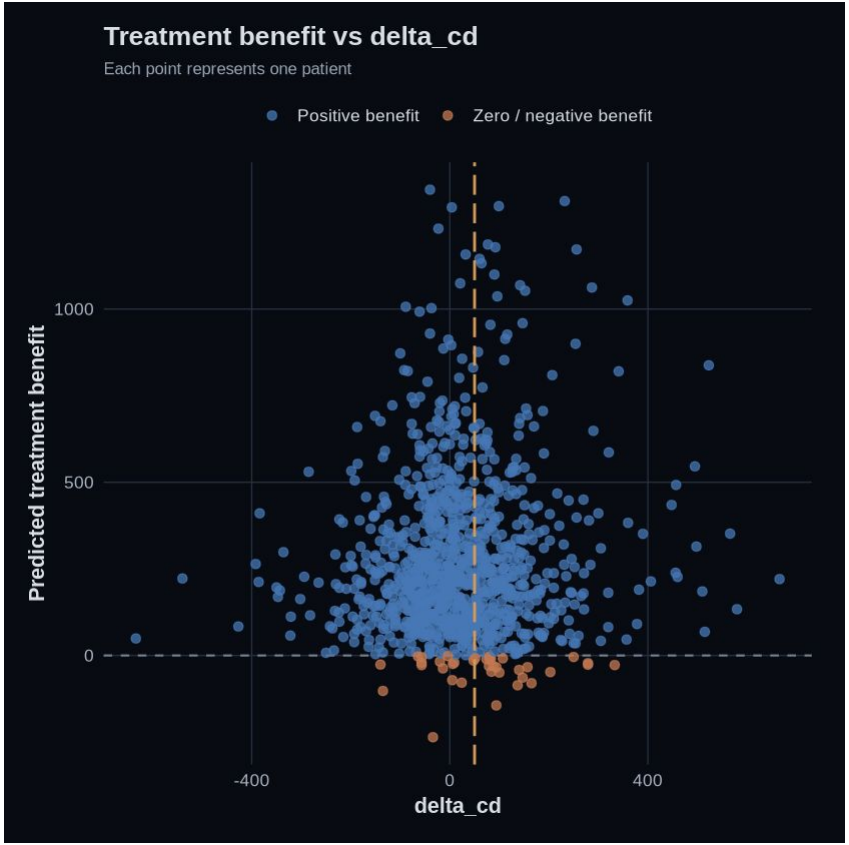
A negative result

This doesn't mean it's a good model – as you see, the model sorted 1017/1054 as high benefit from treatment → not so helpful to us.

As you'll see from the plot, it's not like all patients responded well either:
Delta CD (Change in T-cell count) was ~0 in almost all these patients

Currently, the model has been fit on a useful, diverse set of baseline variables. It has not identified a clear subset.

Quantifying it:
ATE: 281.78 (SD 140.77)



NB

Conclusions

One key assumption at the starting :

- Restricting the analysis to only arms 0 and 1
- This allowed for a clear treatment effect
- It seems the HTE analysis wasn't able to pick out a group Where adding *dd* treatment significantly improved their outcomes

How else could we go about it?
1 +3

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Arm	**ZDV 600**	**ZDV (600mg) + dd (400mg)**	**ZDV (600mg) + ZAL (2.25mg)**	**dd (400mg)**
Patient Count	532	522	524	561

NB

Did I go too simple with the model?

One key assumption at the starting :

- Restricting the analysis to only arms 0 and 1
- This allowed for a clear treatment effect
- It seems the HTE analysis wasn't able to pick out a group Where adding *dd* treatment significantly improved their outcomes
-

Arm #	0	1	2	3
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NB